

Instrument For X-Ray Lithochemistry will be able to determine the elemental composition of materials to a finer scale. With its X-ray fluorescence spectrometer, this instrument essentially makes detection and analysis of the surface materials much more detailed compared to the previous instruments. The other spectrometer is the Scanning Habitable Environments with Raman and Luminescence for Organics and Chemicals (SHERLOC). Using an ultraviolet laser, it's a finer spectrometer which will be used to detect organic compounds and also carry out fine-scale mineralogy. The imaging instruments and spectrometers makes it easier to study rock and soil composition. They could potentially provide evidence of microbial life if organic compounds are detected.

Studying the layers of rock formed on the Martian crust provides insight into past records of the environment on how they were formed and how they changed through time. This will be made possible with the Radar Imager for Mars' Subsurface Experiment (RIMFAX). A centimeter-scale resolution of the geological structure of the crust and the subsurface will be recorded using this ground-penetrating radar.

The Mars Environmental Dynamics Analyzer (MEDA) contains several sensors working together to provide us with a lot of data on the climatic conditions. These include temperature, pressure, wind speed and direction, relative humidity, and dust size and shape.

Coming to one of the crucial instruments that will eventually enable humans to go to Mars is the Mars Oxygen ISRU Experiment (MOXIE). Mars'

FACTS

- **Launch:** July/August 2020
- **Launch vehicle:** Atlas V-541
- **Total weight:** 531,000 kilograms (including full fuel-capacity and spacecraft)
- **Launch location:** Cape Canaveral Air Force Station in Florida, USA
- **Landing:** February 2021
- **Duration:** One Mars year (about 687 Earth days)

atmosphere is about 96 per cent carbon dioxide. If this instrument works, we will not have to worry about producing Oxygen on Mars.

WHERE IS IT LANDING?

Mars is the second-smallest planet in the solar system (after Mercury), but it still has a surface area of about 144.8 million sq km – compare that to earth's land surface area of 148.3 million sq km, and you understand why picking the right landing site is important. Extensive workshops have been conducted to look at the mission objectives, safe landing zones and maximising the rover's exploring ability, and has been narrowed down to three sites. The three landing sites are called Columbia Hills, Jezero Crater and Northeast Syrtis. Columbia Hills/Gusev Crater could potentially harbour an ancient hot spring, Jezero Crater holds a dried-up Martian lake, and Northeast Syrtis is an interesting location that has a shield volcano right beside an impact crater. All three sites are on the Martian equator, and

the rover will probably explore all three if all goes well.

THE FUTURE?

Another interesting mission scientists at NASA are working on is Mars InSight (Interior Exploration using Seismic Investigations, Geodesy and Heat Transport). The primary objective of the lander is to study and understand the process

of how rocky planets formed billions of years ago. The lander will investigate beyond the surface of the planet by studying the tectonic activity and meteorite impacts. This mission will give us an insight about the formation of planets in the solar system including our own. Initially slated for a 2016 launch, the InSight mission has been delayed for a 2018 launch due to a vacuum leak in one of its instruments.

The dream is to drop off astronauts on the red planet by the 2030s, because humans explore a lot better than robots or rovers. Sadly, NASA's chief of human spaceflight, William H. Gerstenmaier, recently announced that they don't have enough funding to "put a date on humans on Mars". It's going to be a long time before the world can once again sit glued to screens as they watch a human go farther than ever before in the bid to explore our solar system, and eventually the galaxy. It's just sad that

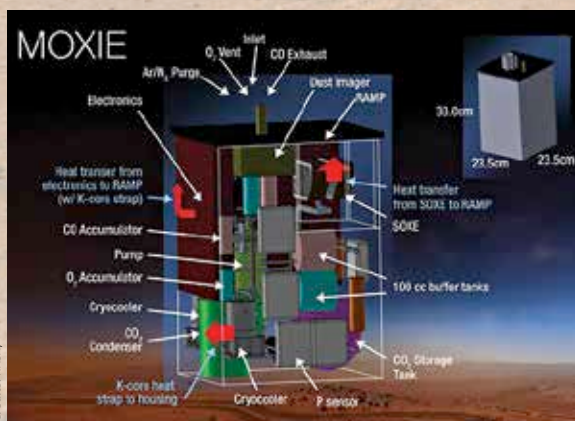


(Credits: NASA/JPL-Caltech)

Mars InSight mission to be launched in 2018

petty politics and a lack of international collaboration are holding us back from exploring space.

Hopefully the Mars 2020 rover will be a success, and we will be able to conclude that Mars did indeed harbour life in the past (or maybe it still does?). If microbial life, or the traces of it from long ago, is actually discovered, perhaps that will be the final humbling our species needs, so that we can get our act together and start working together to preserve life instead of fighting amongst ourselves like squabbling siblings. Exciting times ahead, and Mars might still have important secrets to reveal. We can't wait for the Mars 2020 mission! 🚀



(Credits: NASA)

The Mars Oxygen ISRU Experiment (MOXIE) will convert carbon dioxide from the atmosphere to oxygen